

SOFTWARE REQUIREMENTS & TEAM PLAN

AIIA Clinical Trials Dashboard

A real-time, cloud-based, GCP-compliant Clinical Trial Management System (CTMS) for Ayurveda research — with CDISC/FHIR-interoperable data, role-based KPIs, and integrated ethics, regulatory (CTRI / NDCT Rules 2019) and pharmacovigilance tracking.

FastAPI + PostgreSQL

React + TypeScript + Tailwind

CDISC / FHIR Interoperability

CTRI & NDCT Rules 2019

ML Risk Prediction

RBAC + Audit Trail

Ayush · ML

Hrishikesh · Backend

Aryan · Integration

Meet · Compliance

Swaroop · Frontend

Vedika · Frontend

1

Project Overview

What the system does, and the standards it must meet

CORE PURPOSE

- Manage Ayurveda clinical trials end-to-end: trials, sites, participants, visits and clinical data
- Give every role (Sponsor, PI, Coordinator, Ethics, Regulatory, Monitor) a live, role-based dashboard
- Track safety events and pharmacovigilance in real time
- Export data in CDISC and FHIR formats for interoperability with external systems

REGULATORY & TECHNICAL STANDARDS

- **CTRI** registration & status tracking built into the regulatory module
- **NDCT Rules, 2019** compliance for ethics review workflow and documentation
- **CDISC** (DM, AE, VS, EX domains) and **FHIR** (ResearchStudy, Patient, Observation) export
- Deployed on **Google Cloud Platform (GCP)**, following its security & data-residency guidance

Real-Time

WebSocket/Redis-backed live KPI updates on the dashboard

GCP-Compliant

Cloud SQL (PostgreSQL), Cloud Run/GKE, IAM-based access control

Interoperable

CDISC + FHIR export endpoints for external regulatory systems

Auditable

Immutable audit trail on every clinical & compliance change

2

Final Team Structure

One clearly-owned module per person — no overlap, no gaps

MEMBER	TEAM	PRIMARY RESPONSIBILITY
Ayush	ML + Backend	ML/AI risk-prediction model, ML API, and backend support for the intelligence layer
Hrishikesh	Backend	Database design, authentication, RBAC, and core CRUD APIs (trials, sites, participants, visits)
Aryan	Backend + Frontend	Clinical/safety backend APIs, plus frontend-backend integration lead
Meet	Backend	Ethics, regulatory (CTRI), document management, audit trail, and FHIR/CDISC export
Swaroop	Frontend	Main dashboard, design system, trial management and site management UI
Vedika Sakharkar	Frontend	Participants, visits, clinical data (eCRF), and safety/adverse-event UI

Why Aryan sits in both teams: he is the only member spanning backend and frontend, so he is designated the **API + Frontend Integration Lead** — owning the clinical/safety backend and making sure every frontend screen is actually wired to a real API, rather than static UI.

System Architecture

How the modules connect end-to-end



WHO WORKS WITH WHOM

Hrishikesh ↔ Everyone

Owns database + auth that every other module depends on

Aryan ↔ Frontend + Backend

Bridges Swaroop & Vedika's UI to Hrishikesh/Meet's APIs

Meet ↔ Hrishikesh & Swaroop

Compliance DB design + Ethics/Regulatory/Audit UI

Ayush ↔ Swaroop

Feeds ML insights and risk cards into the main dashboard

Per-Member Requirements (SRS)

Modules, key APIs, deliverables and viva statement for each member

Ayush

AI/ML Engineer + ML Backend Developer

ML + Backend

MODULES

- Trial recruitment risk prediction
- Site performance risk analysis
- Safety signal / risk flagging
- Data preprocessing & model evaluation

DELIVERABLES

- ML dataset + preprocessing pipeline
- Trained model + evaluation report
- ML API + documentation
- AI insight cards for the dashboard

KEY API ENDPOINTS

```
POST /api/ml/recruitment-risk
POST /api/ml/site-risk
POST /api/ml/safety-risk
GET /api/ml/insights/{trial_id}
```

IMPORTANT BOUNDARY

- AI flags risk for human review — it never makes autonomous clinical decisions

"I built the intelligence layer that flags recruitment, site and safety risk for human review — not autonomous clinical decisions."

Hrishikesh

Backend Lead — Database + Authentication

Backend

MODULES

- PostgreSQL schema (users, trials, sites, participants, visits)
- Authentication: login, JWT, refresh tokens
- RBAC across 9 roles (Admin, Sponsor, PI, Coordinator...)
- Dashboard aggregation APIs

DELIVERABLES

- PostgreSQL schema + SQLAlchemy models
- Auth & RBAC modules, DB migrations
- Core + dashboard APIs, API tests

KEY API ENDPOINTS

```
POST /api/auth/login
GET /api/trials POST /api/trials
GET /api/participants/{id}
GET /api/dashboard/overview
```

SECURITY

- Password hashing, CORS, input validation, rate limiting, SQL-injection protection

"I built the secure backend foundation — database, authentication and role-based authorization that every other module relies on."

Aryan

API + Frontend Integration Lead

Backend + Frontend

MODULES

- Clinical data (eCRF, vitals, labs)
- Adverse events & pharmacovigilance workflow
- Protocol deviations
- Frontend service layer (Axios / TanStack Query)

DELIVERABLES

- Clinical, AE, PV & deviation APIs
- services/ layer connecting all frontend screens
- Error handling, loading states, integration tests

KEY API ENDPOINTS

```
POST /api/clinical-data
POST /api/adverse-events
PUT /api/pharmacovigilance/{id}
POST /api/protocol-deviations
```

PHARMACOVIGILANCE FLOW

- Reported → Under Review → Causality Assessment → Follow-up → Resolved

"I own the clinical/safety data layer and connect all frontend apps to FastAPI — making sure workflows are functional, not just static UI."

Meet

Compliance & Governance Backend Developer

Backend

MODULES

- Ethics submissions & review workflow
- Regulatory / CTRI tracking (NDCT Rules 2019)
- Document management with strict versioning
- Immutable audit trail; FHIR & CDISC export

DELIVERABLES

- Ethics + regulatory APIs, CTRI tracker
- Versioned document store, audit log
- FHIR & CDISC export endpoints

KEY API ENDPOINTS

```
POST /api/ethics/submissions
GET /api/regulatory
GET /api/fhir/export/{trial_id}
GET /api/cdisc/export/{trial_id}
```

CRITICAL RULE

- Documents are versioned, never overwritten (v1.0 → v1.1 → v1.2); audit logs are read-only

"I built the compliance layer — ethics, regulatory/CTRI tracking, versioned documents, audit trail and FHIR/CDISC interoperability."

Swaroop

Lead UI/UX + Dashboard Developer

Frontend

MODULES

- Design system (colors, typography, components) — React, Tailwind, shadcn/ui, Recharts
- Main layout: topbar, sidebar, content area
- Main dashboard: KPIs + charts
- Trial management & site management UI

DASHBOARD KPIS

- Active Trials, Participants, Active Sites
- Adverse Events, Pending Approvals
- Expiring Documents, Protocol Deviations

DELIVERABLES

- Design system + responsive layout
- Dashboard, Trials UI, Trial 360 view, Sites UI

"I designed and built the main frontend experience — dashboard, trial management, site management and the reusable design system."

Vedika Sakharkar

Clinical Workflow Frontend Developer

Frontend

MODULES

- Participant management + participant profile / timeline
- Visit management (calendar, list, per-participant views)
- eCRF clinical forms — React Hook Form + Zod
- Adverse event & pharmacovigilance UI

CLINICAL TIMELINE STAGES

- Screening → Consent → Enrollment → Day 7 → Day 14 → Day 30

DELIVERABLES

- Participants + profile pages, visit calendar
- eCRF + adverse event forms, form validation

"I built the clinical workflow frontend — participants, visits, eCRFs and safety events — focused on simple, structured data entry."

Responsibility Matrix

Every feature, one lead, clear support roles

Feature	Ayush	Hrishikesh	Aryan	Meet	Swaroop	Vedika
ML / AI Risk Engine	●		●		●	
Authentication		●	●			
RBAC		●	●			
Database	●	●	●	●		
Trials		●	●		●	
Sites		●	●		●	
Participants		●	●			●
Visits		●	●			●
eCRF			●			●
Clinical Data			●			●
Adverse Events			●			●
Pharmacovigilance			●			●
Ethics				●	●	
Regulatory / CTRI				●	●	
Documents				●	●	
Audit Trail				●	●	
FHIR / CDISC Export				●	●	
Dashboard		●	●		●	
UI Design System					●	●
Frontend Integration			●		●	●
Testing (module-wise)	●	●	●	●	●	●

● Primary owner (Lead) ● Support / Integration

6

6-Week Development Plan

Building complete, working modules in each phase — not disconnected pieces

W1

Foundation

Goal: project skeleton is live with auth working end-to-end

Hrishikesh: DB schema, auth, RBAC, FastAPI setup **Swaroop:** Design system, layout, sidebar, dashboard skeleton

Vedika: Participant UI, visit UI skeleton **Aryan:** API service architecture, clinical data models

Meet: Ethics/regulatory DB design, document architecture **Ayush:** Dataset, ML problem definition, initial model

W2

Core CTMS

Goal: Login → Dashboard → Trial → Site → Participant → Visit flow works

Build out Trials, Sites, Participants and Visits — full CRUD + UI, connected end-to-end

W3

Clinical + Safety

Goal: Participant → Visit → eCRF → Adverse Event → Safety Review flow works

Build eCRF, clinical data capture, adverse events and pharmacovigilance workflow

W4

Compliance

Goal: ethics, regulatory and document workflows are fully auditable

Build Ethics, Regulatory/CTRI, Documents (with versioning) and Audit Trail

W5

AI + Interop

Goal: ML risk insights appear live on the dashboard; data exports validate

Build ML → Risk Analysis → Dashboard pipeline, plus FHIR and CDISC export

W6

Demo Ready

Goal: stable, demo-ready build — no new features

Bug fixing, API integration, security/RBAC checks, responsive design, loading & error states, demo data, deployment, final presentation

7

The One Rule for This Team

Divide by complete modules, not by "frontend person / backend person"

Every member should be able to say, in one sentence, exactly what they own:

Ayush — "I built the AI risk-analysis layer." • **Hrishikesh** — "I built the secure backend foundation, database and RBAC." • **Aryan** — "I built clinical/safety APIs and integrated frontend with backend." • **Meet** — "I built ethics, regulatory, document, audit and interoperability modules." • **Swaroop** — "I built the main dashboard, trial management and site-management experience." • **Vedika** — "I built the participant, visit, eCRF and clinical workflow experience."